

Open Chemical Data in the Age of AI



Noel O'Boyle

**Chemical Biology Resources
Team Leader**

25 Aug 2026, ACS Fall 2026, Chicago
Herman Skolnik Award Symposium in honour of Antony Williams



Simple

Structure

Advanced

ChemSpiderman



Description

ChemSpiderman is the host of ChemSpider and populates the database with structures of interest from blogs, articles and the Chemistry World at large

Company name ChemZoo

Email

Phone

Website <http://www.chemspider.com/blog>

Street address

City

Province/state

Postal/ZIP code

Country

United States of America

Data source classification

404: page not found.

We can't find the page you tried to access. It may have been moved or deleted.

[ChemSpider home](#)

Bibliometricking J Cheminf

Just for fun, a week or two ago I decided to check out who are the most prolific authors in J. Cheminf. I downloaded the TOC in Endnote format from the journal website, and analysed it with the short script below.

There were 295 authors in total (after merging of I think a single duplicate), and all of the authors with 2 or more papers are as follows (by no. of publications, then reverse alphabetical by surname):

14 Peter Murray-Rust
8 Stephen Bryant
8 Evan Bolton
7 Sam Adams
6 Egon Willighagen
6 Joe Townsend
6 Sunghwan Kim
5 Andreas Zell
5 Antony Williams
5 Andreas Jahn
5 Georg Hinselmann
4 David Wild
4 Christoph Steinbeck
4 Noel O'Boyle
4 Geoffrey Hutchison
3 Tim Vandermeersch
3 Henry Rzepa
3 Lars Rosenbaum
3 Matthias Rarey
3 Stefan Kramer
3 David Jessop
3 Nina Jeliazkova
3 Marcus Hanwell
3 Nikolas Fechner
3 Peter Ertl

“Nostalgic prologue”



PMR awarding Blue Obelisks to Andy Lang, Daniel Lowe, Tony Williams, Matt Todd

2014-07-14 Jean-Claude Bradley Memorial Symposium, Cambridge, UK

Credit: Nina Jeliazkova on Twitter

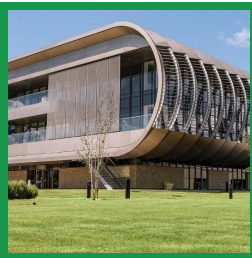
<https://x.com/10705013/status/488672553586937856>

2012-09-03 “Noel O’Blog”

<https://baoilleach.blogspot.com/2012/09/bibliometricking-j-cheminf.html>

(J. Cheminf. was established in 2009)

The European Molecular Biology Laboratory



EMBL-EBI

Bioinformatics



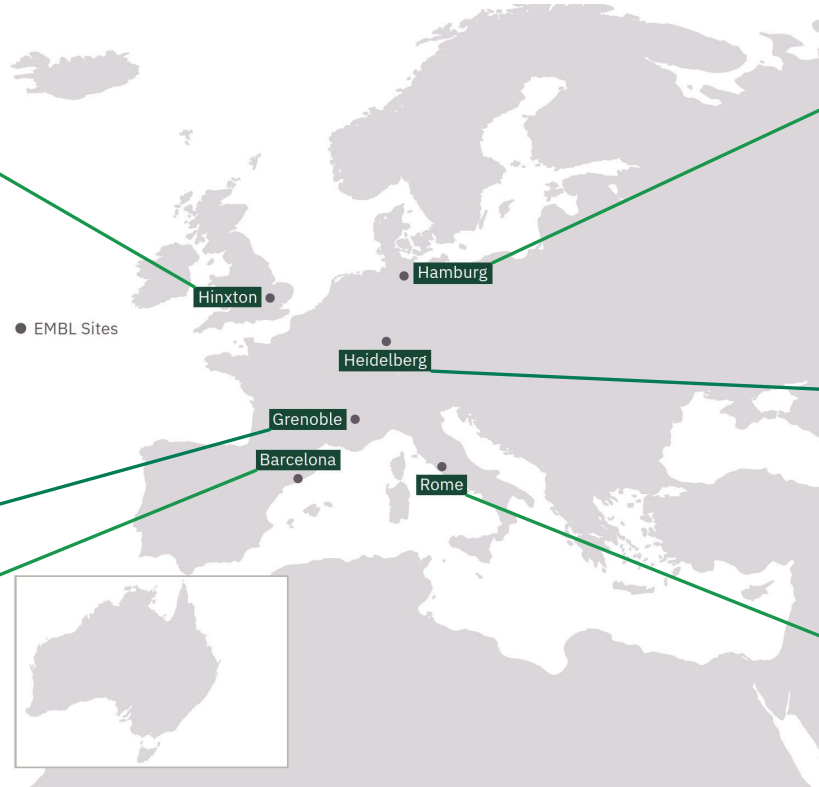
Grenoble

Structural biology



Barcelona

Tissue biology
and disease
modelling



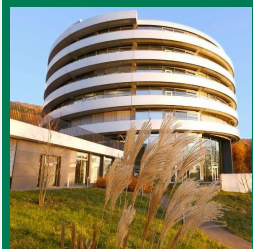
Hamburg

Structural biology



Heidelberg

Life sciences



Rome

Epigenetics
and neurobiology



Chemical Biology Resources @ EMBL-EBI



ChEMBL

*Bioactivities
from literature
and deposited*

*Drug/candidate
information*



SureChEMBL

*Chemical and
biomedical data
from patents*



UniChem

*Link databases
via InChI*



ChEBI

Chemical ontology

*Chemical
registration db for
biological dbs*



OPSIN

*Convert IUPAC
names to structure*

Chemical Biology Resources @ EMBL-EBI

Curated databases



ChEMBL

*Bioactivities
from literature
and deposited*

*Drug/candidate
information*



ChEBI

Chemical ontology

*Chemical
registration db for
biological dbs*

Automatically-populated databases



SureChEMBL

*Chemical and
biomedical data
from patents*



UniChem

*Link databases
via InChI*

(Not a database)

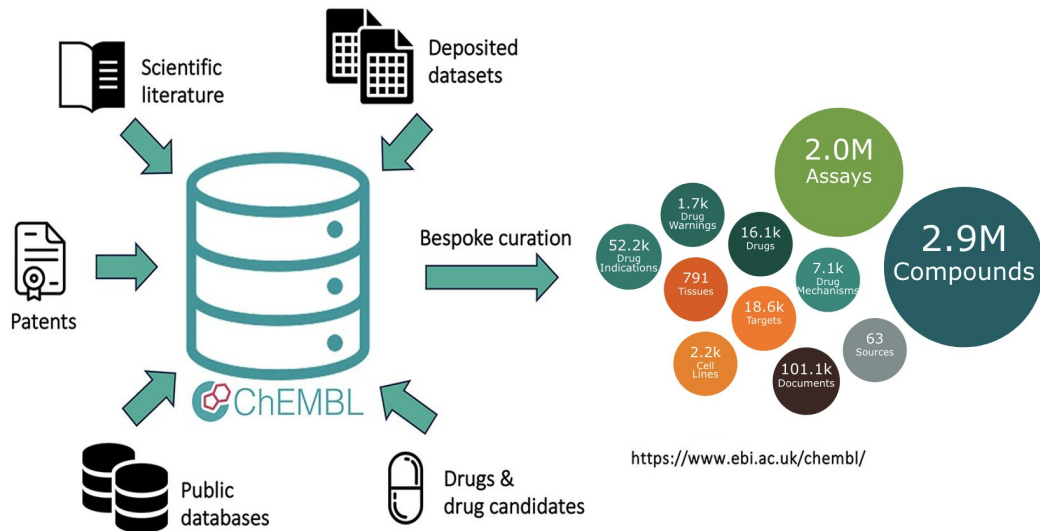


OPSIN

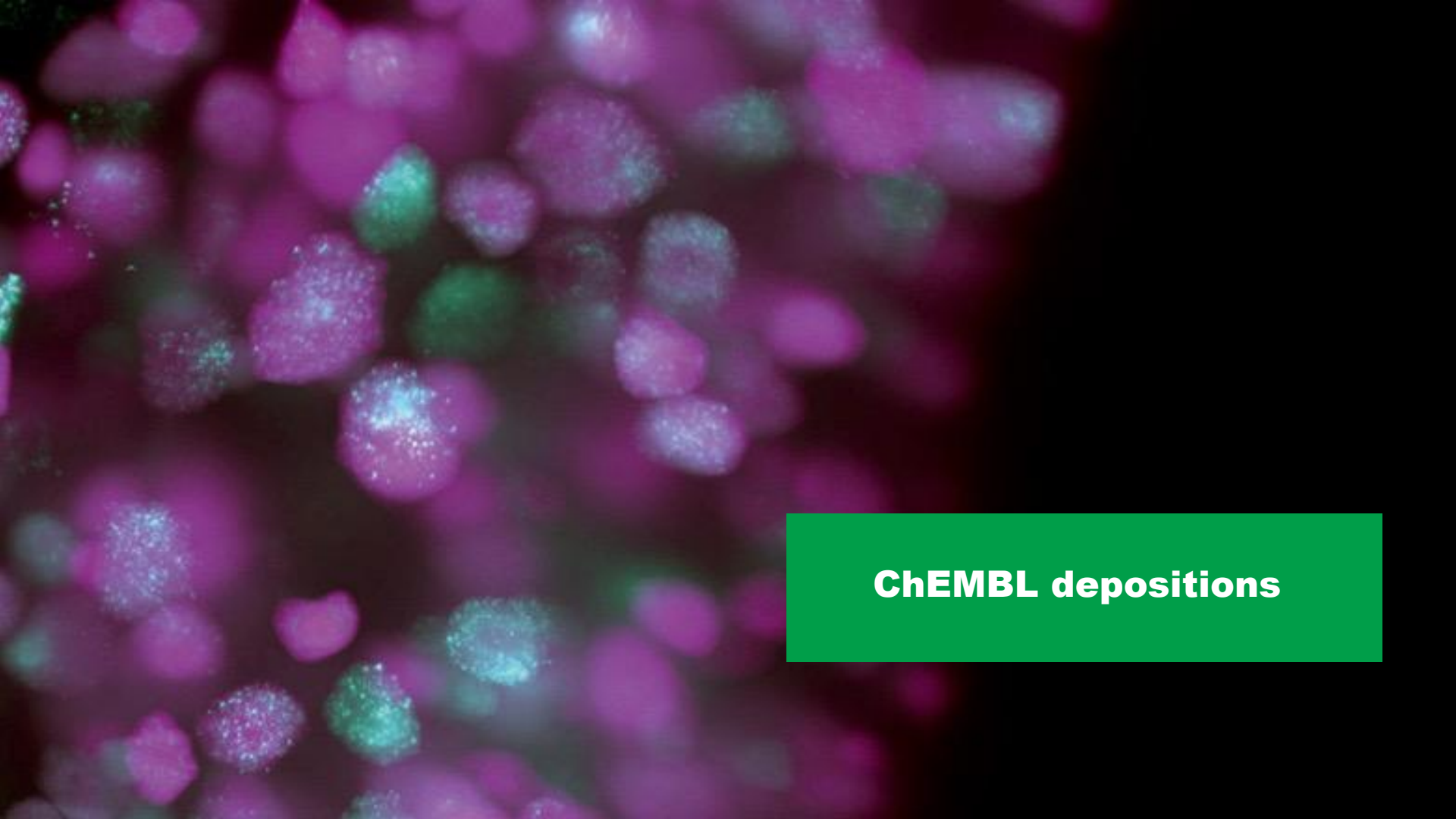
*Convert IUPAC
names to structure*

ChEMBL database

- Launched in 2009 with 440K structures, now with 2.9M
- Chemical structures in ChEMBL arise from a wide variety of data sources
- We extract bioactivity data from the **scientific literature**
 - Comprehensively for seven core medicinal chemistry journals
- **Drug and clinical candidate data** is manually curated from USAN, INN and EMA, and clinical trials.
- **Deposited datasets** have grown in size, and now account for ~60% of bioactivities



Zdrazil B. et al NAR 2024 52:D1180-D1192. doi: 10.1093/nar/gkad1004.
PMID: 37933841.



ChEMBL depositions

What people
imagine...



"Upload"



Curation of high-quality ML-ready data requires resource

What actually happens...



- Work with depositor to harmonise data format
- Request missing fields
- Extend ChEMBL data model
- Discussions on new data types
- Group data appropriately into assays and documents
- Fix typos, unusual characters

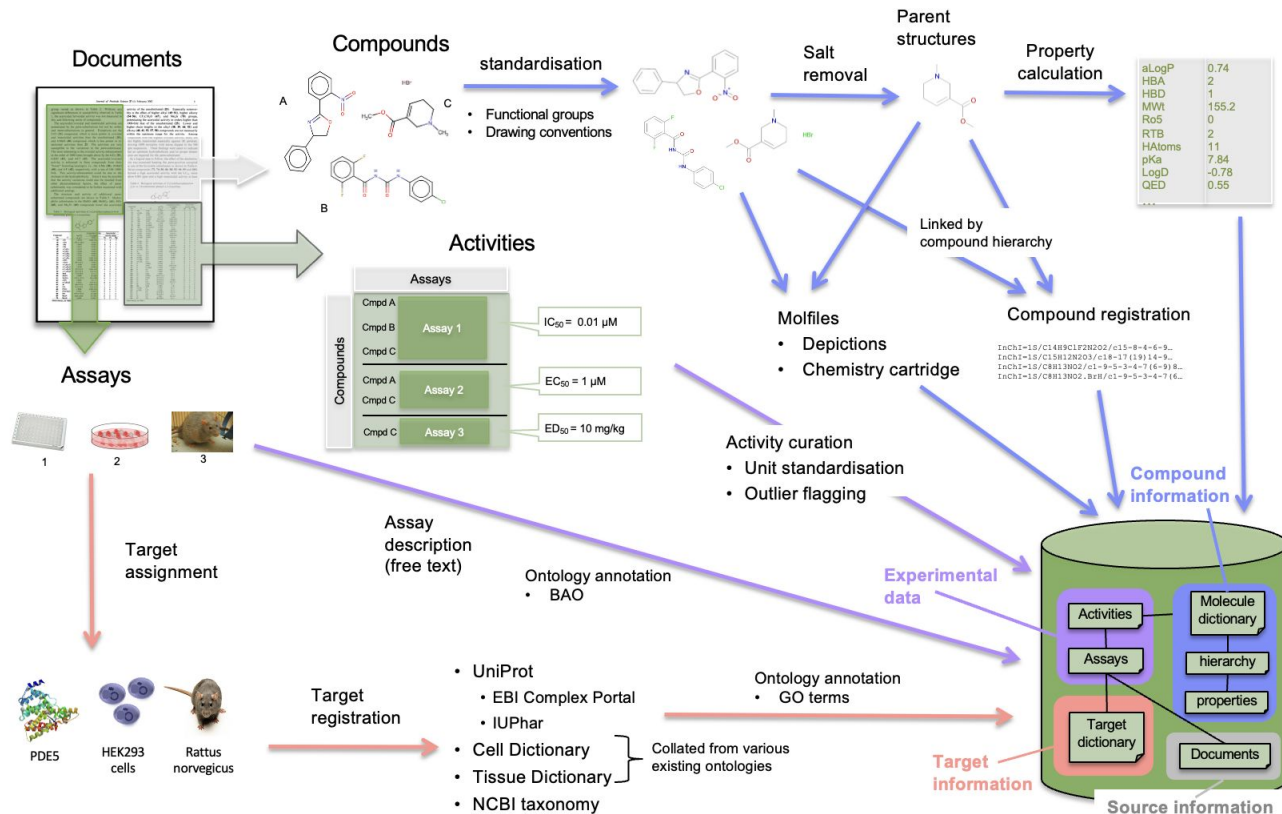


- Scientific validity: compare results to provided abstract (conversion errors)
- Encourage additional information to help with interpretability
- Check assignation of targets, and map to dictionaries
- Map other biological entities or variants present in assay to dictionaries
- Check for duplicates (and fix)
- Check assignation of compound names
- ...



> 1.5 FTE / year over 5 years

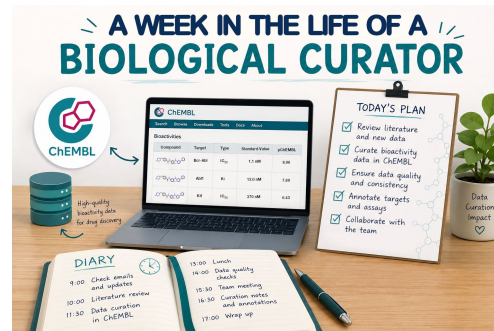
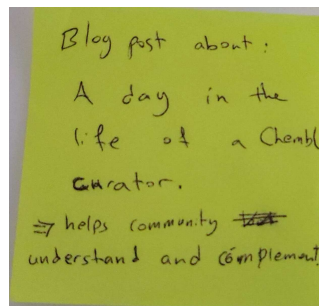
Data quality & data curation in ChEMBL



Bento, A.P. et al. **An open source chemical structure curation pipeline using RDKit.** *J. Cheminform.* 2020, 12, 51.

https://github.com/chembl/ChEMBL_Structure_Pipeline

Papadatos G. et al. **Activity, assay and target data curation and quality in the ChEMBL database.** *J. Comput. Aided Mol. Des.* 2015, 29, 885.



Emma Manners



Salesia Werner

Excerpt from “Monday”

...In the afternoon, it's time for focused work on the database. Today, I'm running checks on a newly deposited dataset. Before it can be incorporated into ChEMBL, I look for anything that seems unusual using Jupyter notebooks:

- Do the reported activities make sense in the context of the assay?
- Have experimental details such as targets, cell lines and tissues been captured correctly?
- Are there any duplicate assays or activity records?
- Do compound names and structures conflict with existing ChEMBL records?

Any issues found are captured in GitLab for discussion with the team. By the end of the day, I hope I've caught everything that might cause issues later and I can log off knowing that the dataset is one step closer to becoming part of ChEMBL.

Inter-curator agreement across mapped fields

Agreement between Sally & Emma

(Assay count)



Variant ID



Assay type



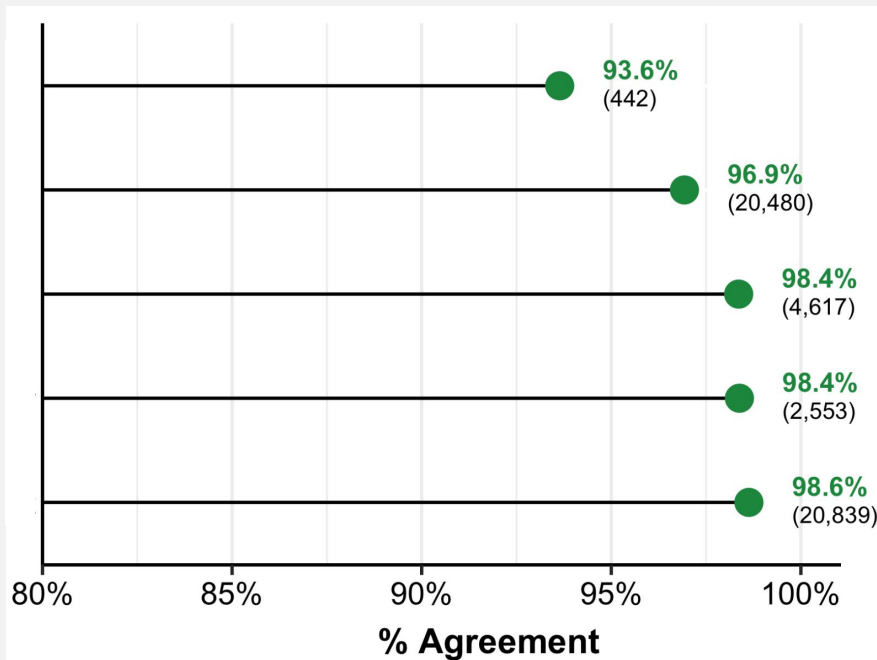
Tissue ID



Subcellular fraction

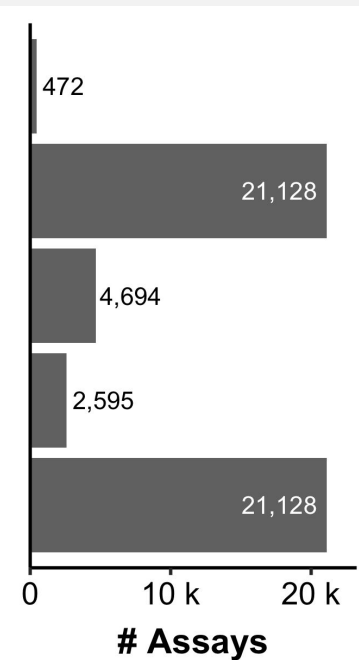


Target ID (tid)



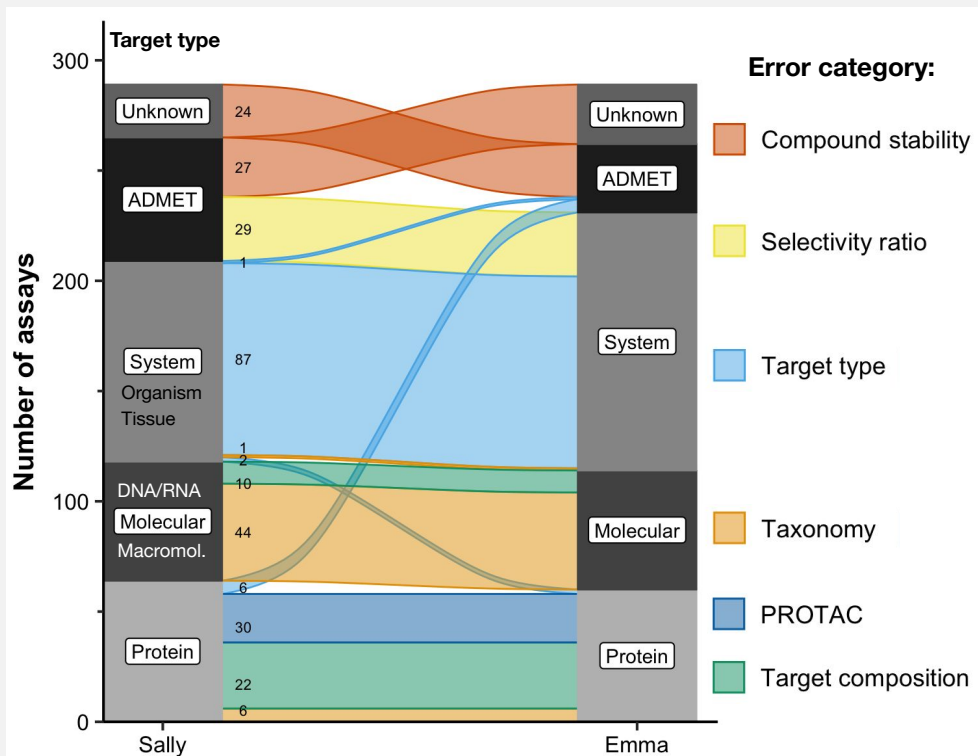
Set size

Total assays compared





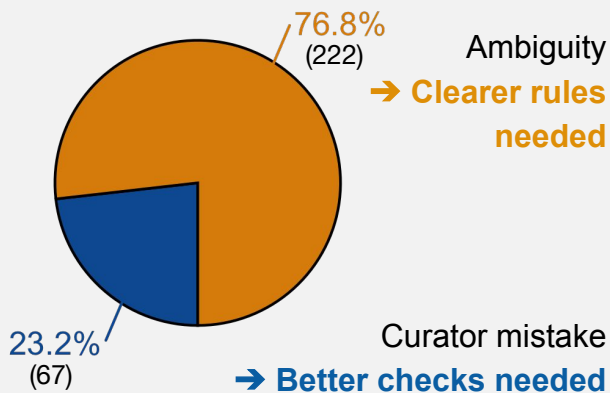
Target ID: Where do disagreements occur?



Example assays:	Sally	Emma
• Drug conversion in PBS	No target	ADMET
• Ratio of EC50 for antiviral activity (Gag A364V mutant / wild type HIV1)	ADMET	Organism (HIV1)
• Effect on CD3+ pan-T cell population in mouse spleen	Tissue (spleen)	Organism (mouse)
• Binding affinity to HER2 G4 DNA	Human	Unknown
• Induction of PD-L1 degradation	Single protein	Interaction
• Inhibition of human CDK2/Cyclin A	Cyclin A2	Cyclin A



Target ID: Where do disagreements occur?

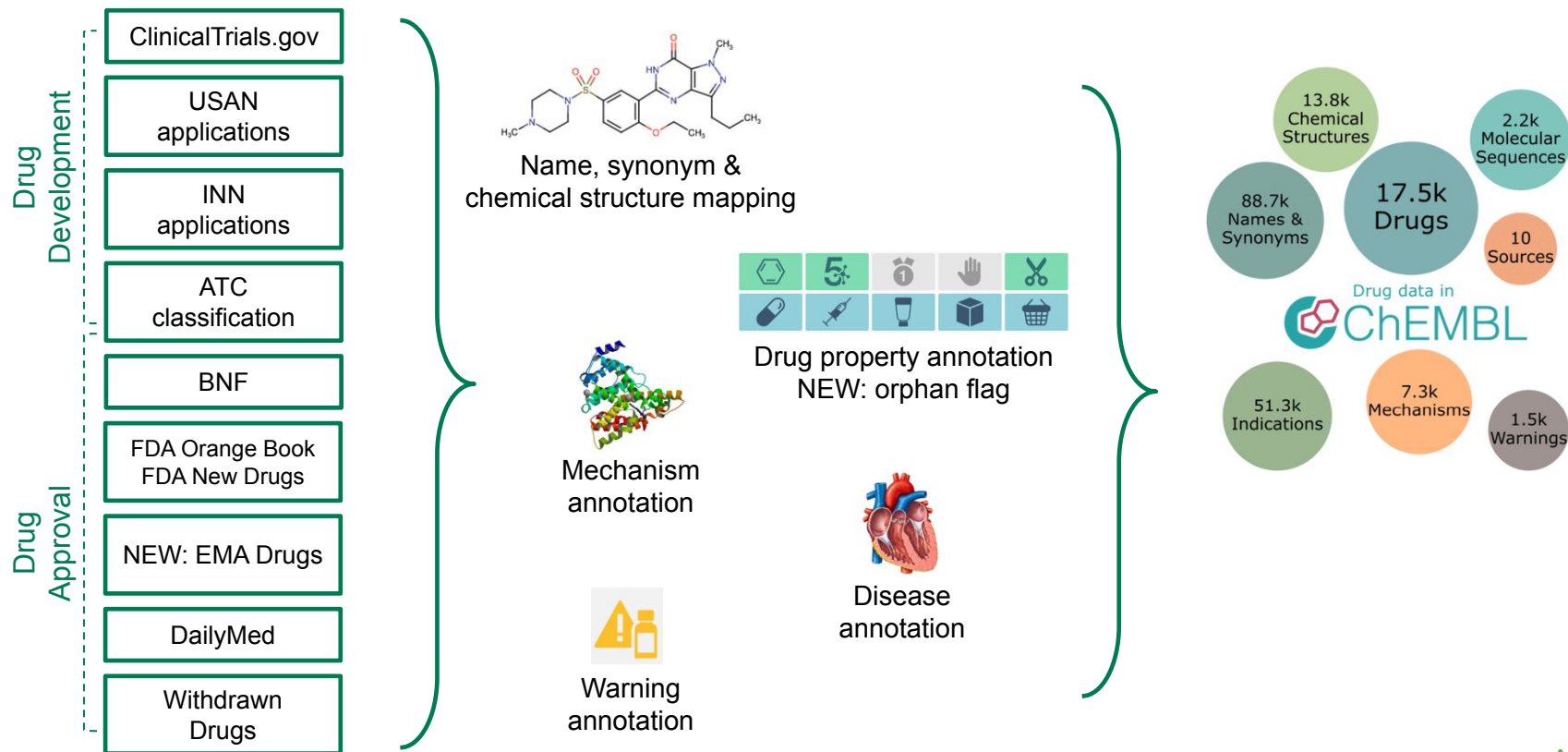


Error category:	Example assays:	Sally	Emma
Compound stability	• Drug conversion in PBS	No target	ADMET
Selectivity ratio	• Ratio of EC50 for antiviral activity (Gag A364V mutant / wild type HIV1)	ADMET	Organism (HIV1)
Target type	• Effect on CD3+ pan-T cell population in mouse spleen	Tissue (spleen)	Organism (mouse)
Taxonomy	• Binding affinity to HER2 G4 DNA	Human	Unknown
PROTAC	• Induction of PD-L1 degradation	Single protein	Interaction
Target composition	• Inhibition of human CDK2/Cyclin A	Cyclin A2	Cyclin A



Drug Data

ChEMBL Drug and Clinical Candidate data



Capturing a drug or clinical candidate

- Several sources such as USAN and INN provide a structure
 - If two sources have the same name but a different structure (or vice versa), this is flagged for curation
 - Typical problems: data source describes the parent compound, while other sources describe the salt; a difference in stereochemistry
- Not all sources provide a structure
 - For ATC and ClinicalTrials.gov, the compound name is matched against existing names/synonyms in ChEMBL, and failing that, against GSRS
 - In the case of GSRS, a check is made for any ChEMBL entries with the same structure, and any inconsistencies are flagged for curation
- Preferred name
 - For approved drugs, the drug name given by the regulator (EMA, FDA, etc.) is assigned
 - Otherwise:
 - Use the USAN name if available
 - Use the INN if available
 - Use the preferred name from GSRS

Data Quality Dimensions

Assessment	What it means	Key Assessment Points	Indicators/Examples
Accuracy	Data reflects real-world biology/clinical reality	Manual curation + community feedback + internal issue tracking ensure correctness	Very low external help-desk drug issues (~8 in 6 months vs 217M API calls); 133 internal issues (120 closed) show active correction process
Completeness	All required data is present for use cases	Major regulators (FDA, EMA) + clinical sources (ClinicalTrials.gov, INN, USAN) integrated	EMA adds only 8% new drugs vs FDA → high coverage overlap; broad annotation of MOA, indication, warnings, properties
Uniqueness	No unnecessary duplicates	Compound registration prevents duplicate structures; manual resolution of edge cases (stereochemistry, naming conflicts)	~0.75% structures manually corrected; synonym duplication reduced over releases (e.g., 2043 → 1843); ongoing reduction of duplicates

Data Quality Dimensions

Assessment	What it means	Key Assessment Points	Indicators/Examples
Consistency	No contradictions across datasets/records	Cross-source validation (FDA/EMA/labels); manual correction of mismatched salt/parent forms and names	~9% EMA-linked drugs required structure/pref_name corrections due to inconsistencies
Timeliness	Data is up to date and refreshed regularly	Each release updates all sources; additions/removals tracked per dataset	~4% of WHO ATC entries changed between releases; increasing use of timestamps for pipelines
Validity	Data conforms to expected formats, ontologies, and ranges	Standardized naming + controlled ontologies (MeSH, EFO); obsolete identifiers regularly remapped	1% pref_name corrections; 11% EFO + 0.05% MeSH identifiers remapped due to ontology updates



Chemical structures

Structure checks

- The ChEMBL structure pipeline was set up with the help of Greg Landrum in 2020. It is the first step when running data checks on any new compound data submitted to ChEMBL.

Checker

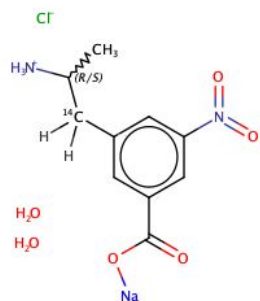
- Empty CTAB
- Polymer
- V3000 MOL file
- 3D coord in MOL file
- Illegal bond type
- Overlapping atoms
- Coordinates all zero
- Crossed bonds in ring
- ...

Standardiser

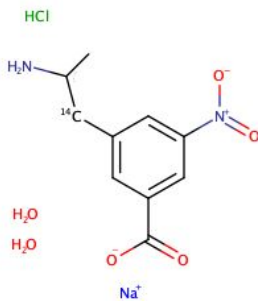
- Clear S Block data
- Kekulise structures
- Fix bad valence
- Fix KO to K+O- & NaO to Na+O-
- NO₂ groups to [N+](=O)[O-]
- NH+Cl- to N and HCl
- Fix wiggly bonds on sp³ C
- Remove explicit Hs
- Straighten triple bonds & allenes
- Standardise sulfoxides
- ...

Salt Stripper (GetParent function)

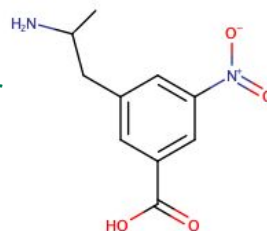
- Salts removed according to ChEMBL salt list
- Solvents removed according to solvents list
- Remove isotopes
- Both components are salts
- Re-neutralise molecule if necessary
-



Standardiser



Salt Stripper



Paper:

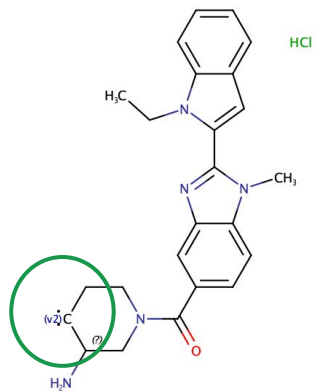
Bento et al. An open source chemical structure curation pipeline using RDKit. *J. Cheminf.* **2020**, 12, 51.

Code:

https://github.com/chembl/ChEMBL_Structure_Pipeline

Structure checks

- Some incorrect structure examples picked up with the structure_pipeline during the release process of ChEMBL36:

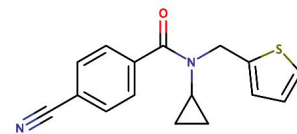
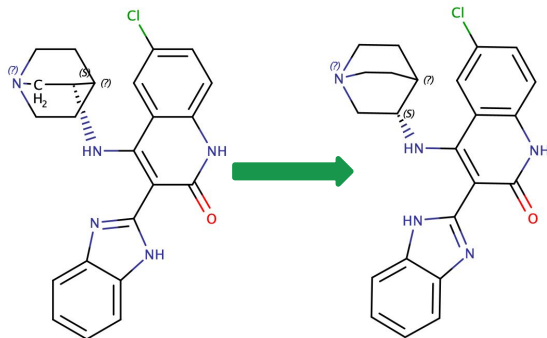


CHEMBL5094982

Penalty score 6, 'molecule has a radical that is not found in the known list'

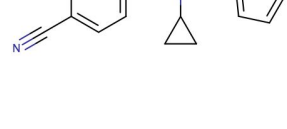
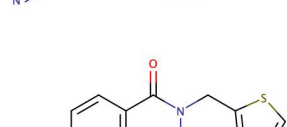
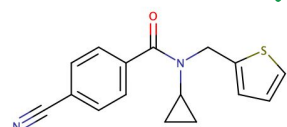
CHEMBL377312

Penalty score 5, 'RDKit_Mol/InChI stereo mismatch'

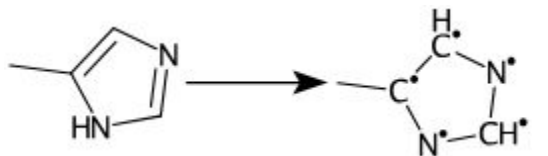
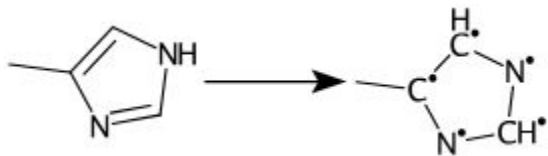


CHEMBL4758943

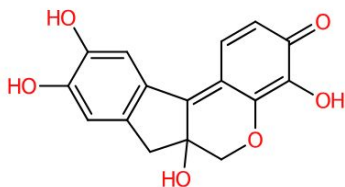
Penalty score 6, 'molecule has six (or more) atoms with exactly the same coordinates'



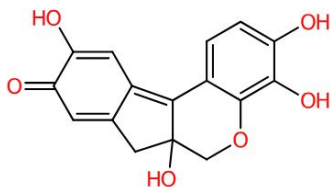
Identifying tautomers using a tautomer hash



C[C]1[CH][N][CH][N]1_1



CHEMBL1730100

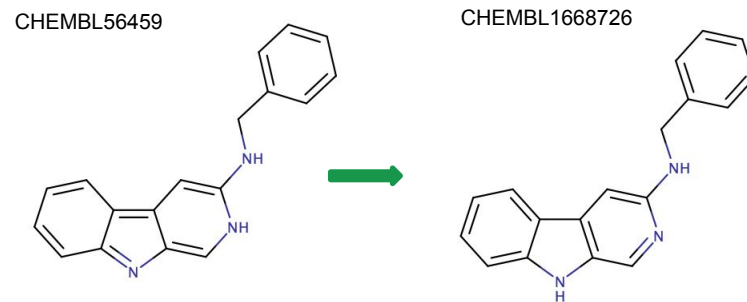
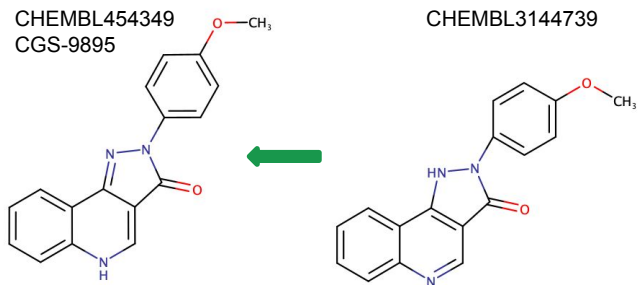


CHEMBL1360563

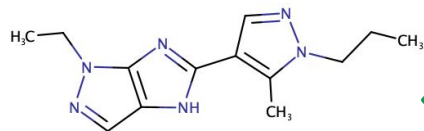
- The 'tautomer hash' can be used to identify additional tautomers beyond those supported by InChI
 - See <https://nextmovesoftware.github.io/molhash> for more details and associated code
 - Limitation: only involves proton shifts on heteroatoms
- Create a canonical SMILES string after setting all the bonds to single bonds, and removing Hs from heteroatoms, and setting charges to 0
- Subtract the original charge from H count and append it to the SMILES string to give the **tautomer hash**
- 550 tautomer pairs were found using this method
 - Mostly 1,4 and 1,5 shifts
 - Example on left is 1,11 shift

Merging tautomer duplicates

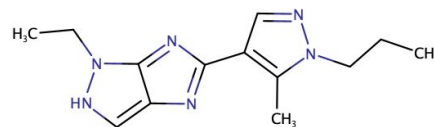
- For the ChEMBL 37 release, 400 tautomers (out of the 550) have been merged
- The remainder will be merged in the upcoming ChEMBL 39



CHEMBL3481372



CHEMBL4527401



Chemical Biology Services



Noel
O'Boyle



Barbara
Zdrzil



Tefvik
Kizilören



James
Webster



Emma
Manners



James
Blackshaw



Marleen de
Veij



Harris
Ioannidis



Salesia
Werner



Nicolas
Bosc



Melissa
Adasme



Ines Smit



Carlos
Moreno



Muhammad
Arsalan



Adnan
Malik



Raymund
Stefancsik



Christine
Withers



Idil Ismail



Marcelo
Almanza

wellcometrust
Strategic Award



Questions about ChEMBL data? Corrections?
Email: chembl-help@ebi.ac.uk

Join the team!

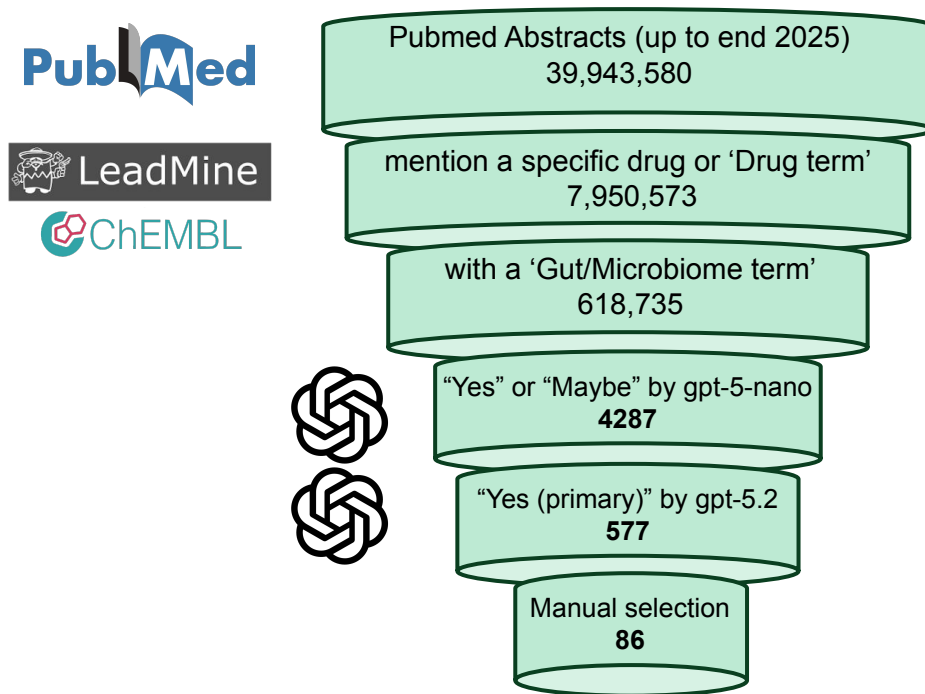
- ChEBI chemistry curator (by Sept 9th 2026)
 - ARISE2 Fellowship opportunity (with AZ, get in touch ASAP by early Sept)
-Coming soon, biological curator, possibly also scientific developer





The Age of AI

Identify papers that describe gut microbiome-mediated biotransformations of drugs



Extraction

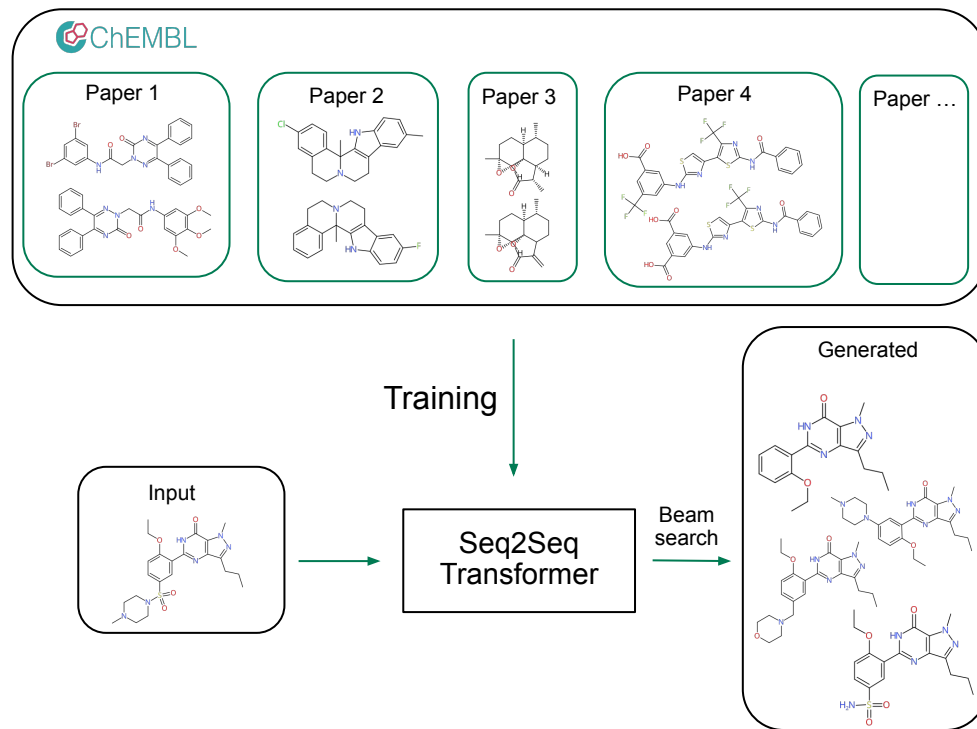


EMBL Heidelberg
Michael Zimmermann
Eleonora Mastrorilli
Mahnoor Zulfiqar

EMBL-EBI
Noel O'Boyle
Barbara Zdrazil
Nicolle Pereira



Generative model for MedChem-similar molecules



PAPER ID: 4537916